

A tensor based reduced order model for parametric interpolation for reactive flows

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Abstract.

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1. Introduction

The development of digital twins for reactive flows is one of the most important steps towards industry decarbonization. Being able to accurately reproduce the behaviour of a system allows to optimize industrial processes, improve energy efficiency and control strategies. Classical computational fluid dynamics (CFD) generates high fidelity simulations of how a system evolves, but due to the fact that is often expensive and time consuming it cannot provide a continuous profile of the behaviour of the system. For this reason algorithms that upsample or super resolute the operational parametric space currently are a growing branch of research. The main goal is that of leveraging the information contained in the high fidelity CFD data to train a model able to generate a simulation of the system under new operating conditions in a fraction of the time. In recent years some techniques based on Singular Value Decomposition (SVD) have been developed to do this [?] [?]. The main underlying idea is to use SVD to decompose the data into linear combination of eigenvectors, often called modes, and then train a machine learning model to estimate the coefficients of the combination for new points.

This work presents a novel reduced order model (ROM) based on the Higher Order Singular Value Decomposition (HOSVD) instead of SVD. Section AAAA presents the current methods used for parametric interpolation and their limitations. Section BBBB describes the proposed method and its advantages. Finally, Section CCCC presents the results of the method on a test case and compares it with existing methods.

2. Problem Formulation

3. Methods

Both methods described are fairly similar from a mathematical point of view. Training a machine learning model directly on CFD data is not feasible due to its extremely high dimensionality. If for example we consider a simulations that refine 100 points for each 3D spatial dimension and 20 thermochemical variables the dimensionality is $\mathbb{R}^{100^3 \times 20} = \mathbb{R}^{2 \times 10^7}$, which would require both a huge dataset and time for any machine learning model. The idea of SVD-based ROMs is to first reduce the dimensionality of the data by reorganizing the CFD simulations into a matrix $\mathbf{M} \in \mathbb{R}^{N_d \times N_s}$, so that the columns of $\mathbf{M}$ are the simulations reshaped as vectors. Applying SVD to $\mathbf{M}$:

$$\mathbf{M} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T \quad (1)$$

where $\mathbf{U} \in \mathbb{R}^{N_s \times N_s}$ and $\mathbf{V} \in \mathbb{R}^{N_d \times N_d}$ are orthogonal matrices, and $\mathbf{\Sigma} \in \mathbb{R}^{N_d \times N_s}$ is a rectangular diagonal matrix with non-negative entries $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r \geq 0$, with $r = \text{rank}(\mathbf{M})$. The columns of $\mathbf{V}$, denoted $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r \in \mathbb{R}^{N_d}$, are the right singular vectors of $\mathbf{M}$ and are referred to as *modes*. Each mode is a vector living in the same space as the original snapshots and captures a coherent spatial structure of the

dataset; the associated singular value σ_j quantifies the contribution of the j -th mode to the overall variance of the data, with the modes ordered such that the first ones retain the most energetically significant structures.

Equation ?? says that each vectorized simulation can be rewritten as linear combination of left eigenvectors with coefficients given by the right eigenvectors. The key assumption for this class of ROMs is that new simulation that one wants to find lie on the span of the left eigenvectors estimated from the data. It is worth noting that this assumption is reasonable given a *steep* enough singular value decay.

Retaining only the $k \ll r$ leading modes, any snapshot $\mathbf{s} \in \mathbb{R}^{N_d}$ can be approximated as an explicit linear combination:

$$\mathbf{s} \approx \sum_{j=1}^k z_j \mathbf{v}_j = \mathbf{V}_k \mathbf{z} \quad (2)$$

where $\mathbf{V}_k \in \mathbb{R}^{N_d \times k}$ collects the first k modes and $\mathbf{z} = (z_1, \dots, z_k)^T \in \mathbb{R}^k$ is the vector of modal coefficients, obtained by projection:

$$\mathbf{z} = \mathbf{V}_k^T \mathbf{s} \quad (3)$$

The truncation rank k is selected by requiring that the retained modes explain a sufficiently large fraction of the total variance:

$$\mathcal{E}(k) = \frac{\sum_{i=1}^k \sigma_i^2}{\sum_{i=1}^r \sigma_i^2} \geq \tau \quad (4)$$

for some threshold τ close to unity, so that the approximation error introduced by the truncation is negligible.

3.1. HOSVD+GPR

The main idea of the algorithm is very similar to

3.2. Hyperparameter Selection

4. Experimental Setup

5. Results

6. Discussion

7. Conclusions